

Technical Debt Document

1. Identified Technical Debts

1.1 Map and Routing Integration Debt

Description:

The application currently uses **Google Maps** for displaying map tiles and **OpenRouteService API** for drawing routes (polylines) between two points. This combination was selected to quickly enable core mapping and routing functionality.

Reason for Debt:

The approach relies on **third-party APIs**, which were chosen for convenience and speed during initial development. However, these external services come with usage limitations, potential costs, and dependency risks.

Impact:

- Continuous dependency on third-party APIs.
- Possible API cost increases or access limitations.
- Integration maintenance challenges.
- Reduced flexibility for local customization or offline support.

Planned Resolution:

Migrate both map rendering and routing functionalities to **Gebeta Maps**, a **local mapping solution** that offers better regional accuracy, lower latency, offline support, and independence from external providers.

Priority: High

Target Resolution: Next release phase after Gebeta Maps API is stable.

1.2 Delivery Confirmation Simulation Debt

Description:

The final **delivery confirmation** process is currently **simulated** to enable development of the **mock receipt page** and demonstrate a working end-to-end user flow. The confirmation action is not yet connected to the live backend API.

Reason for Debt:

The simulation was introduced as a **temporary solution** to unblock frontend development while the backend integration for delivery confirmation is still in progress.

Impact:

- Mock data may not align with live backend responses.
- Delivery status updates are not persisted in the database.
- The demo flow does not fully represent production behavior.

Planned Resolution:

Integrate the delivery confirmation with the **actual API endpoint** to ensure real-time data updates, receipt generation, and accurate delivery tracking.

Priority: Medium

Target Resolution: Next integration sprint.

3. Summary Table

#	Technical Debt	Type	Impact Level	Planned Resolution
1	Map and Routing Integration	Design / Dependency	High	Replace Google Maps & OpenRouteService with Gebeta Maps
2	Delivery Confirmation Simulation	Process / Integration	Medium	Integrate live API for final delivery confirmation

4. Conclusion

Addressing these technical debts will improve **system reliability, maintainability, and independence from external services**. As new features are introduced, similar debts should be tracked, prioritized, and resolved in upcoming development cycles to ensure long-term quality and scalability.